

$$F = G \frac{m_1 m_2}{d^2}$$

$$\phi(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$i\hbar \frac{\partial}{\partial t} \psi = \hat{H} \psi$$

$$F - E + V = 2$$

Economics for Engineers

Module 2 Part 1

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$E = mc^2$$

$$\frac{df}{dt} = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$$

$$dS \geq 0$$

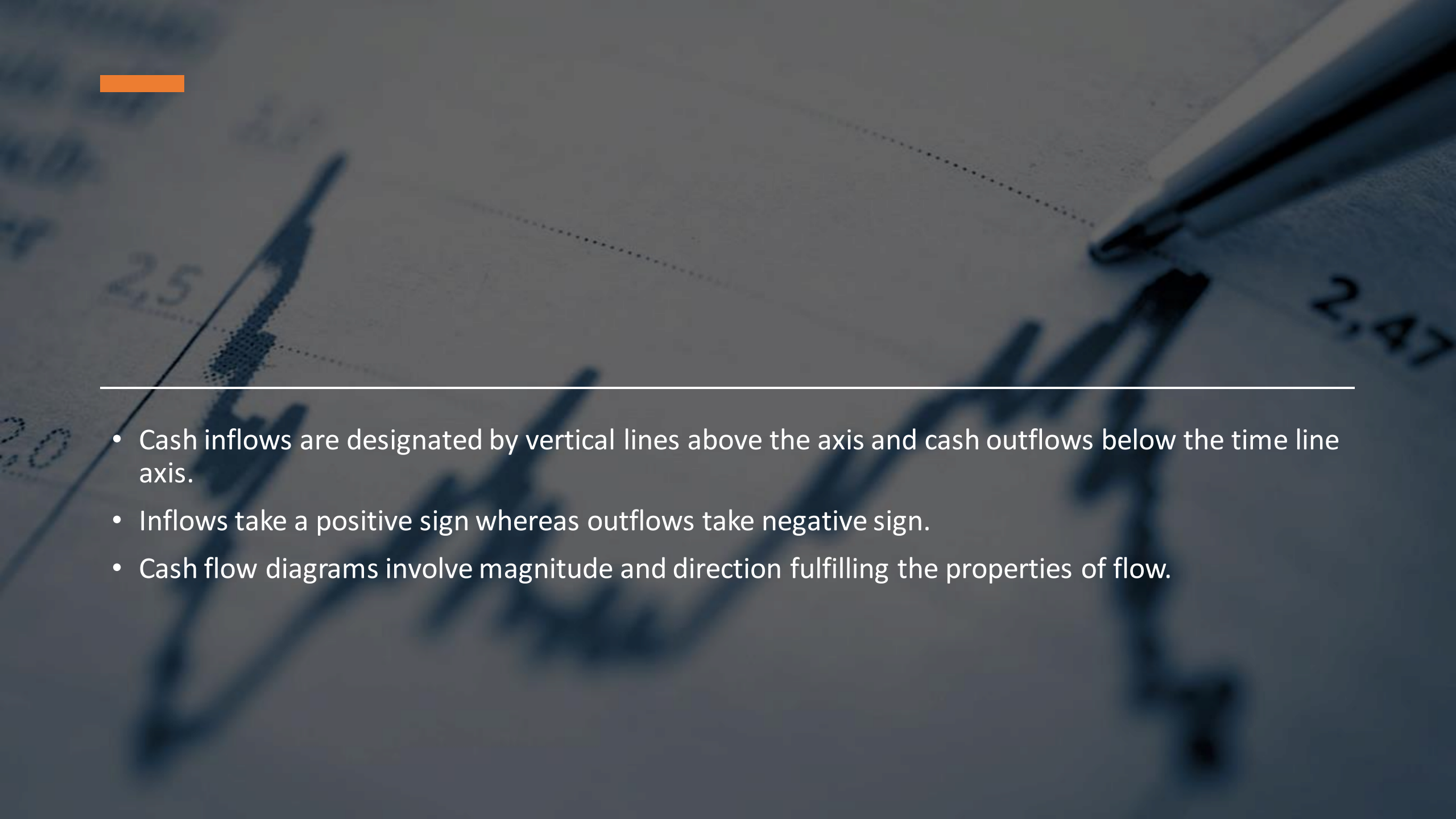
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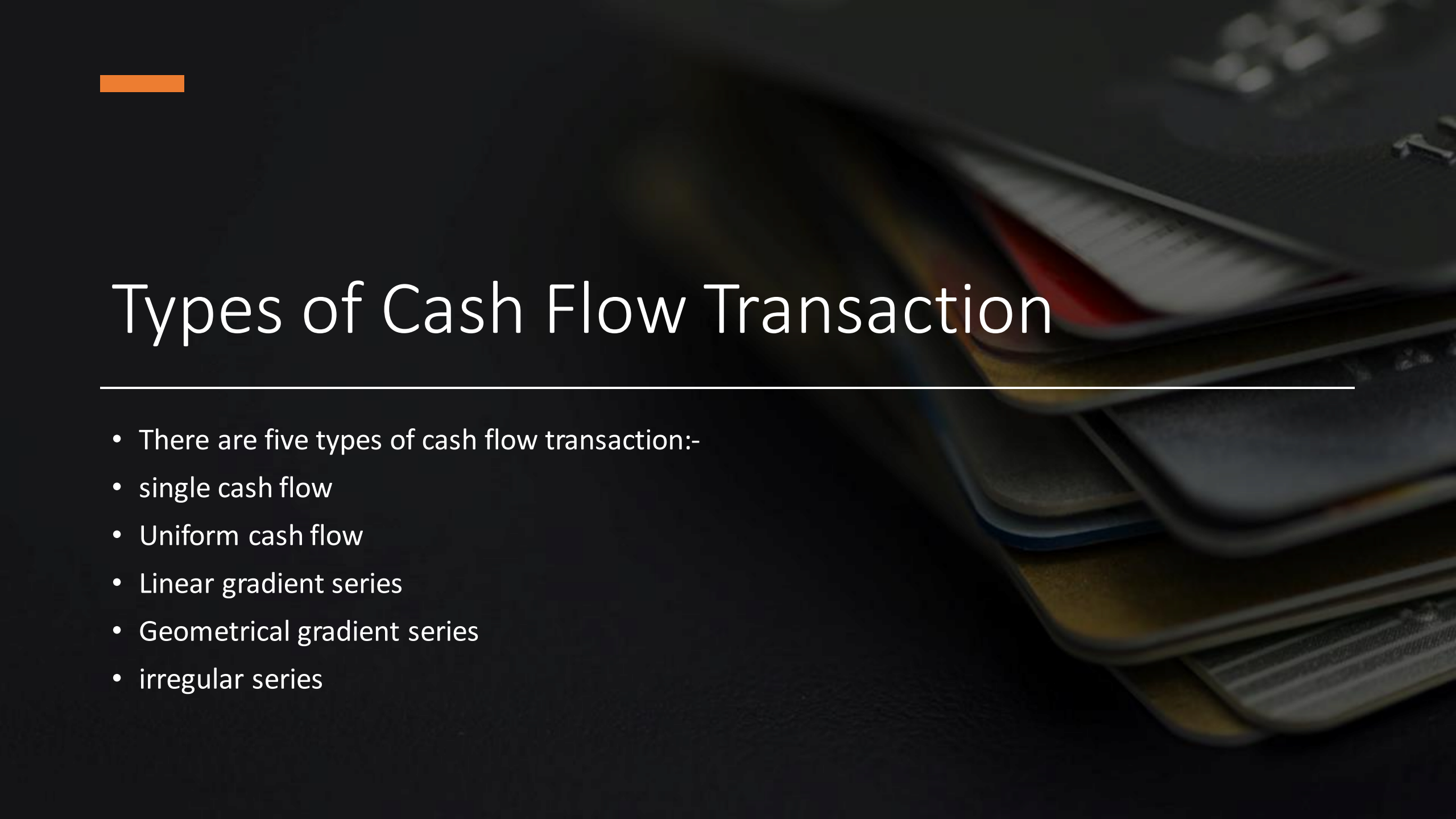
- Cash flow and diagrams
- Time value of money
- Interest rates



Cash flow

- What is cash flow?
- The net amount of cash and cash equivalents being transferred in and out of a company is called cash flow.
- What is cash flow diagram?
- Cash flow diagrams are graphical representation of cash flows (inflows and out flows) along a time line.

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- A background image showing a hand-drawn cash flow diagram on a piece of paper. A pen is visible in the top right corner, and the diagram consists of a horizontal axis with vertical lines representing cash flows. Some numbers like '2.5' and '2.47' are visible on the paper.
- Cash inflows are designated by vertical lines above the axis and cash outflows below the time line axis.
 - Inflows take a positive sign whereas outflows take negative sign.
 - Cash flow diagrams involve magnitude and direction fulfilling the properties of flow.

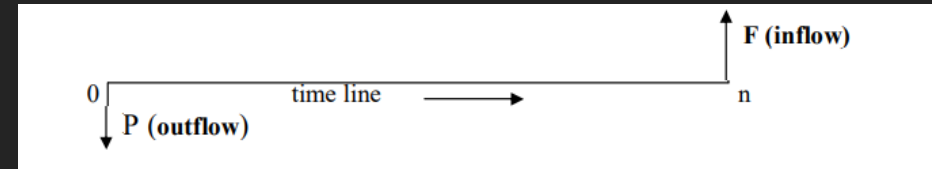


Types of Cash Flow Transaction

- There are five types of cash flow transaction:-
- single cash flow
- Uniform cash flow
- Linear gradient series
- Geometrical gradient series
- irregular series

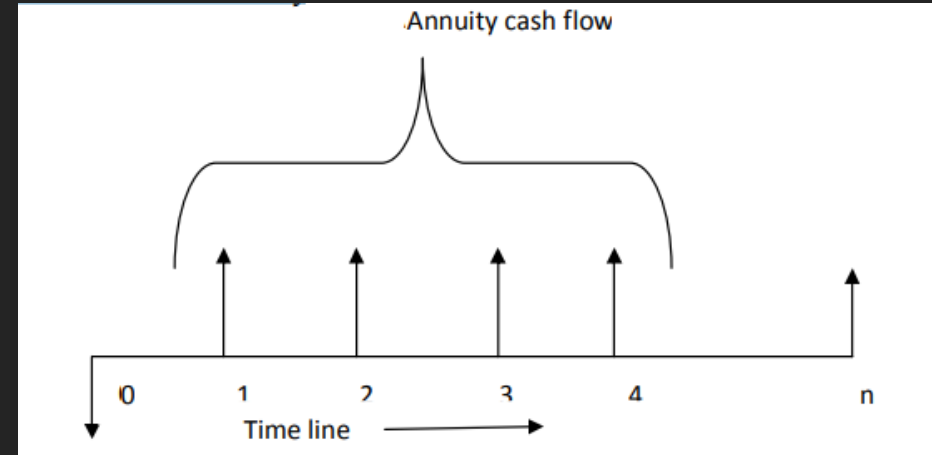
Single payment cash flow

- It involves a single present or future cash flow. The diagram of this is given below.
- In this diagram there is single cash out flow (P) which occurs at '0' time and Cash inflow of (F), which occurs at nth time.



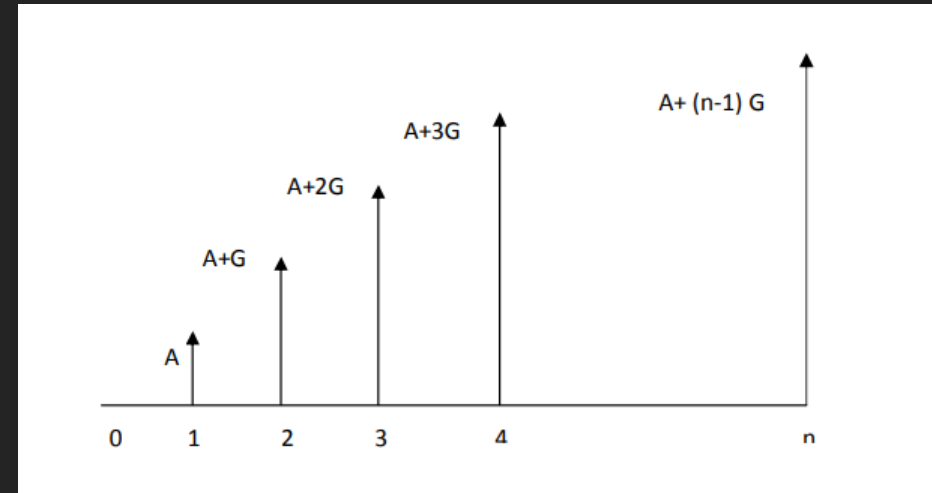
Uniform cash flow

- Here cash flows involve a series of equal amounts occurring at equal interval of time. It is also known as annuity.



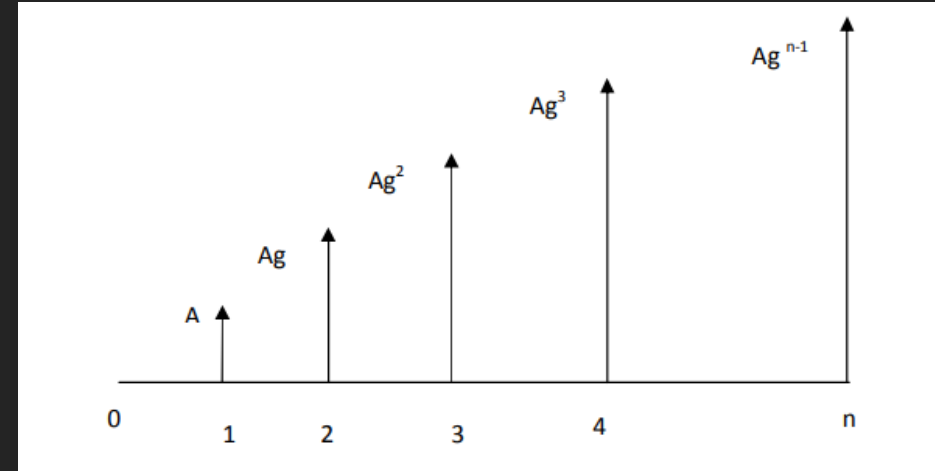
Linear Gradient series

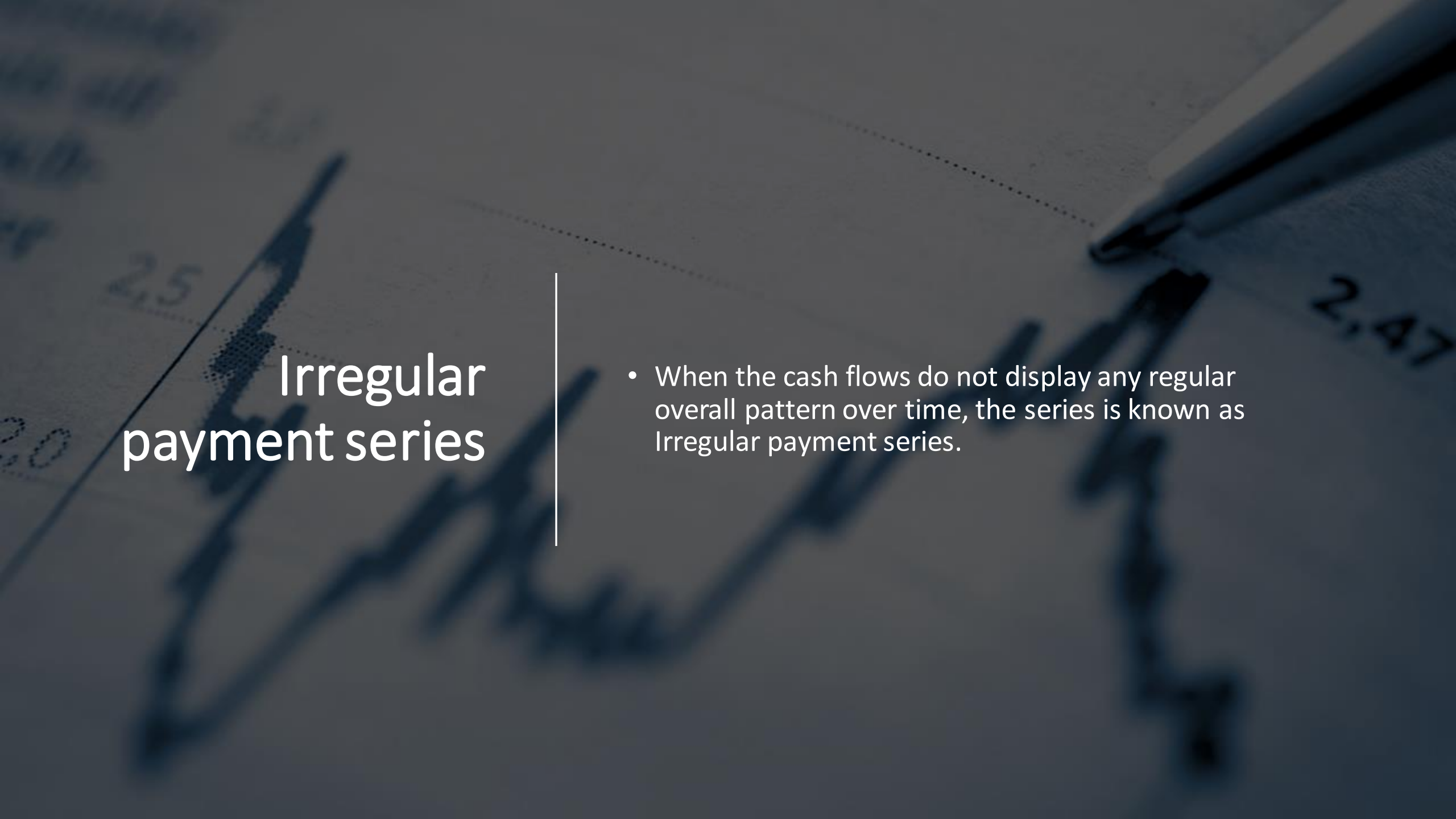
- It is a series of cash flows displaying properties of arithmetic progression series. So the cash flow increases or decreases by a fixed amount.



Geometric Gradient series

- It is a series of cash flows displaying properties of geometric progression series. So the cash flow increases or decreases by a fixed ratio (or percentage).



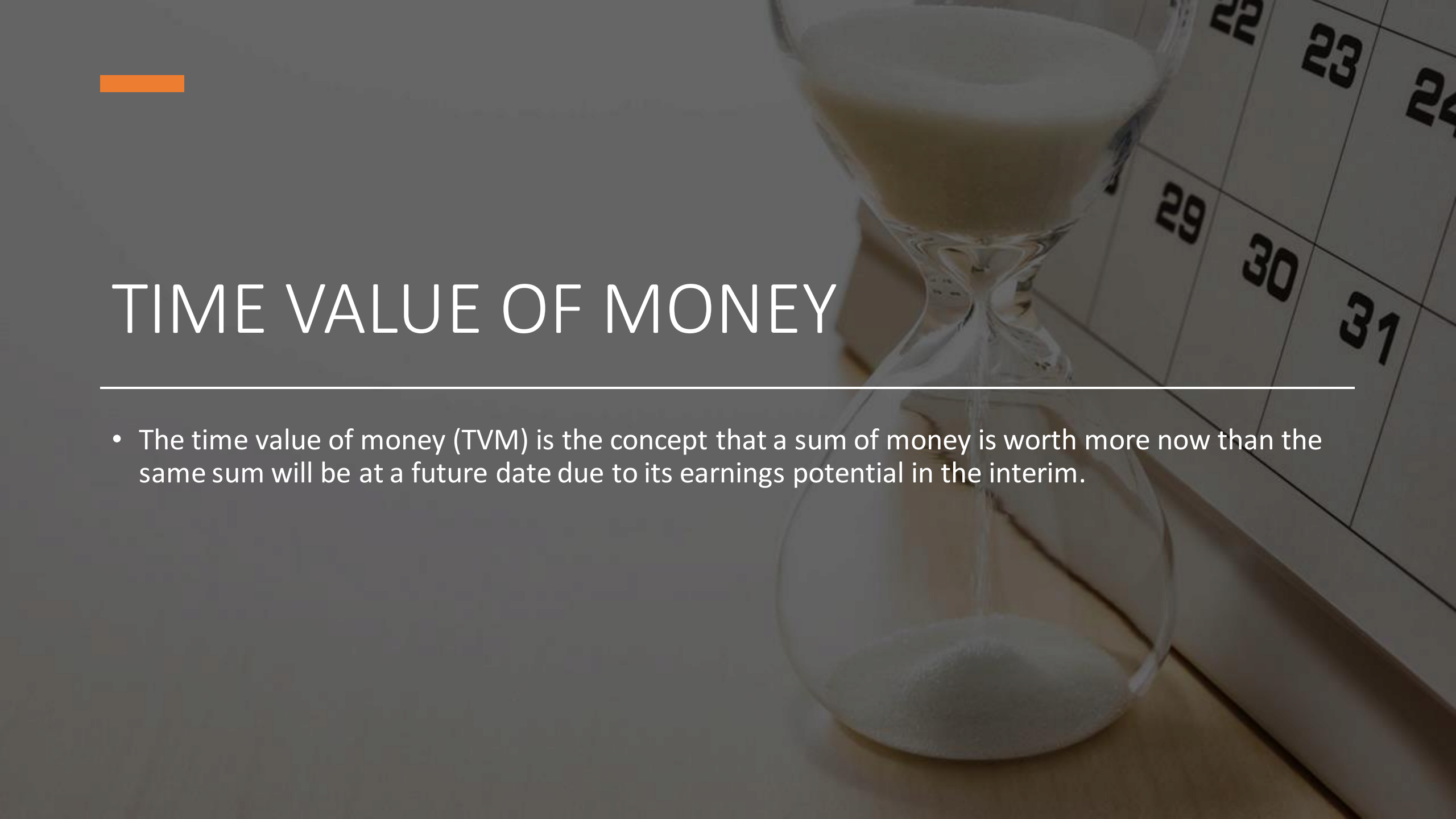
The background of the slide is a dark, blurred image of a pen writing on a document. A line graph is visible, with a dotted trend line and some handwritten numbers like '2.5' and '2.47'.

Irregular payment series

- When the cash flows do not display any regular overall pattern over time, the series is known as Irregular payment series.



TIME VALUE OF MONEY



TIME VALUE OF MONEY

- The time value of money (TVM) is the concept that a sum of money is worth more now than the same sum will be at a future date due to its earnings potential in the interim.

Example:-

- A person has received a prize from a finance company during the recent festival season. But the prize will be given in either of the following two modes.
- Spot payment of \$ 24,72
- \$100 after 10 years from now(this is based on 15% interest rate compounded annually)

Example:-

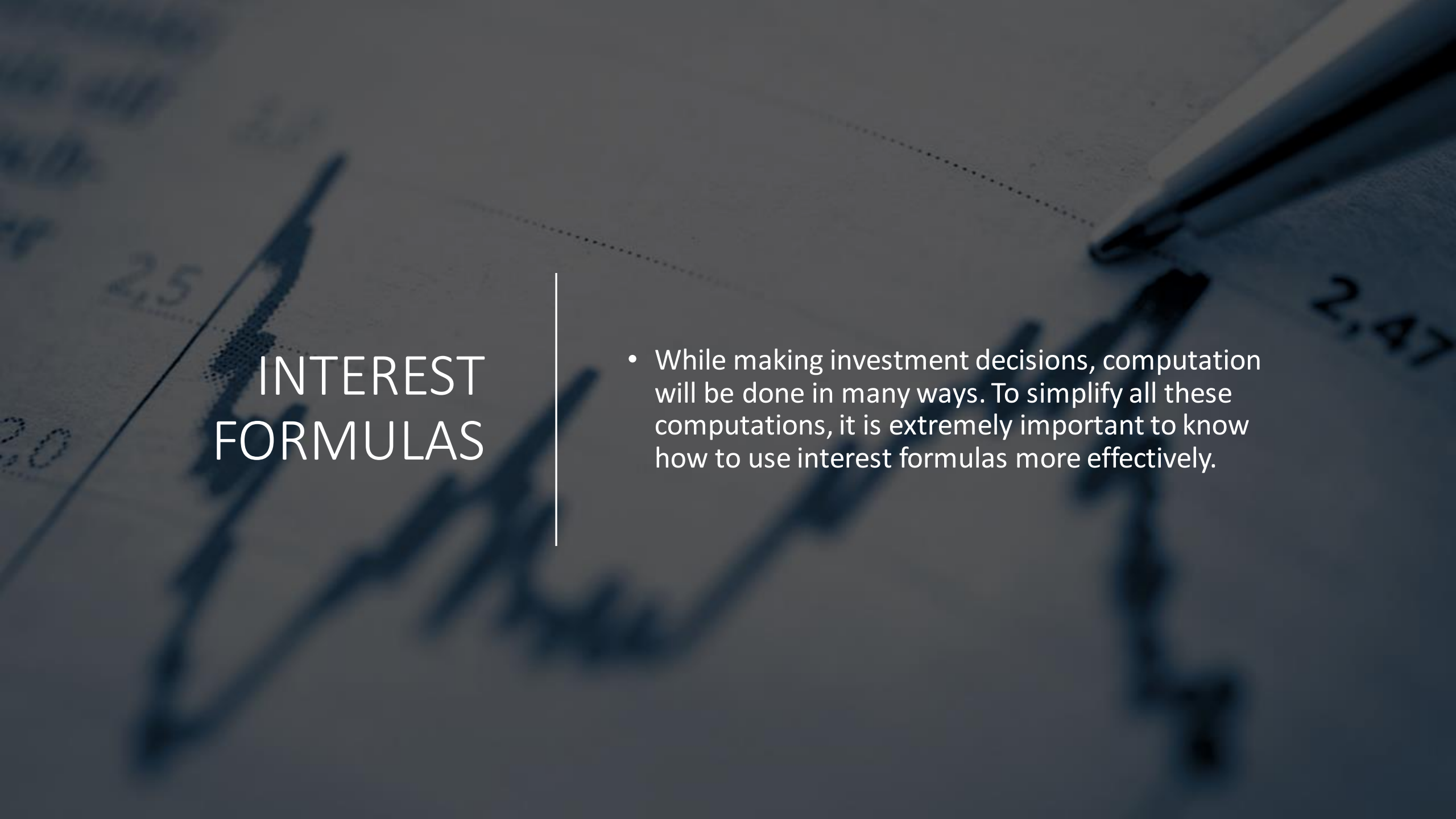
- If the prize winner has no better choice that can yield more than 15% interest rate compounded annually, then it makes no difference whether he receives \$24.72 or \$100 after 10 years.
- This conclusion is derived by calculating value of money across different timelines using “TIME VALUE OF MONEY”.

INTEREST RATES



INTEREST RATES

- Interest rate is the rental value of money. It represent the growth of capital per unit period. Because of interest rate and time factor, the numerical value of money (not purchasing power) multiplies over a period of time.

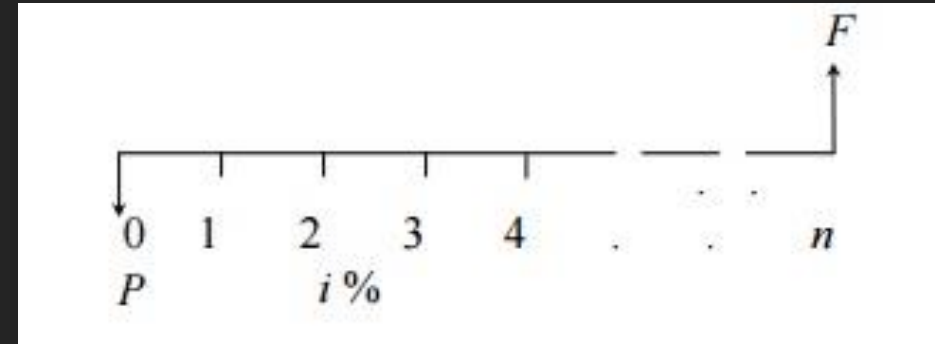
The background is a dark, blurred image of a pen writing on a document. The document contains some faint, handwritten numbers like '2.5' and '2.47'. A vertical white line separates the title from the list item.

INTEREST FORMULAS

- While making investment decisions, computation will be done in many ways. To simplify all these computations, it is extremely important to know how to use interest formulas more effectively.

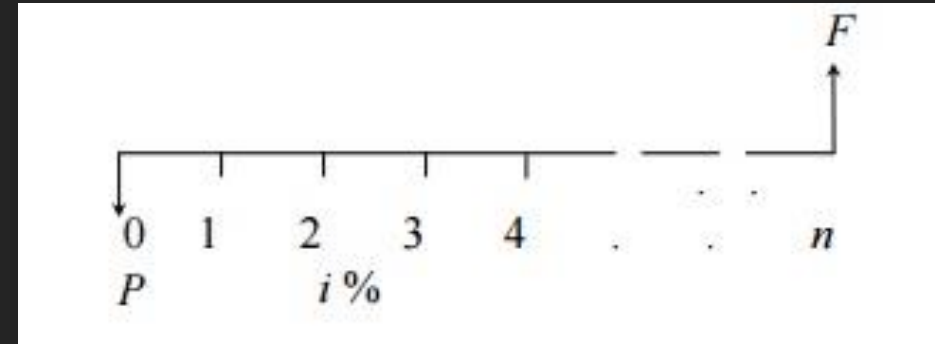
1. Single Payment Compound Amount

- The objective is to find the single future sum (F) of the initial payment (P) made at time 0 after n periods at an interest rate i compounded every period.



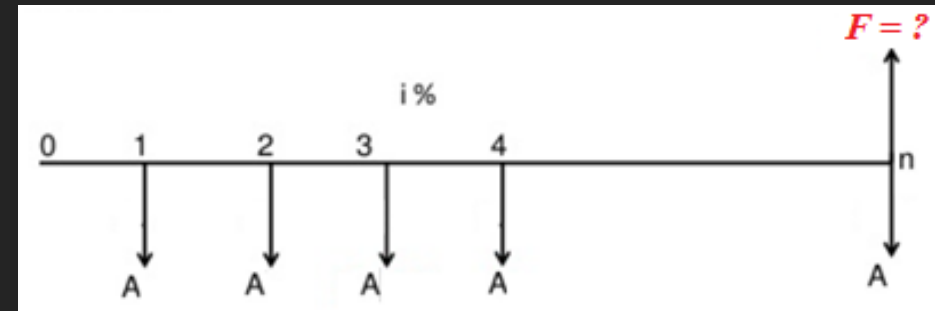
2. Single payment present worth amount

- The objective is to find the present worth amount (p) of a single future sum (f) which will be received after n periods at an interest rate of i compounded at the end of every interest period.



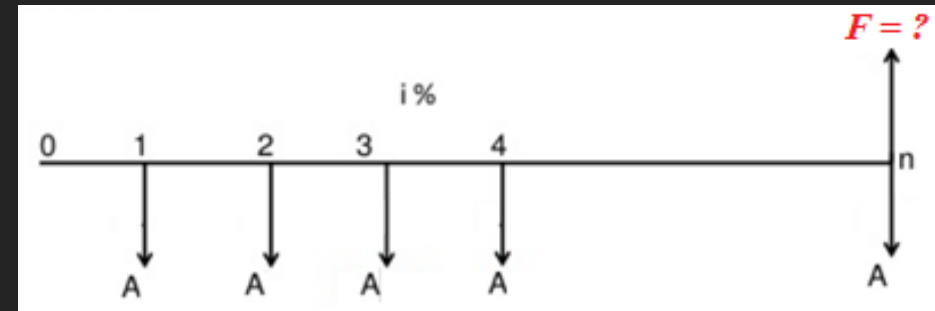
3. Equal Payment series compound amount

- The objective is to find the future worth of n equal payments which are made at the end of every interest period till the end of the n th interest period at an interest rate of i compounded at the end of each interest period.



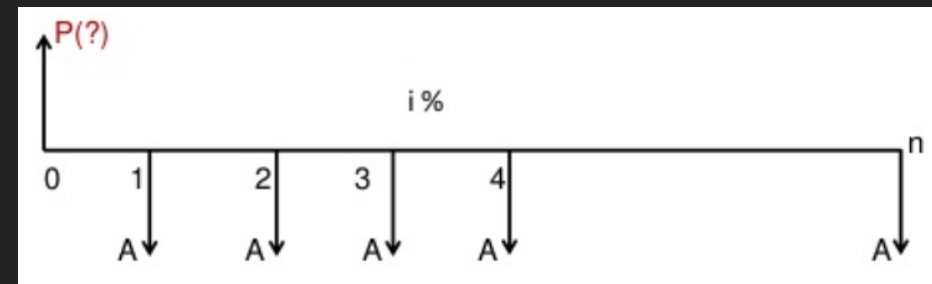
4. Equal Payment series sinking Fund

- The objective is to find the equivalent amount (A) that should be deposited at the end of every interest period for n interest periods to realize a future sum (F) at the end of the n th interest period at an interest rate of i .



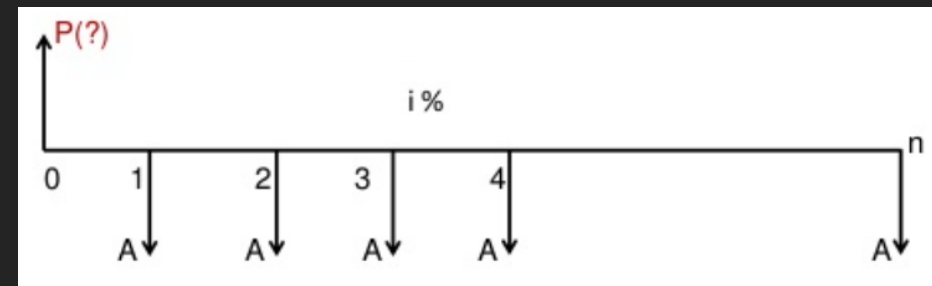
5. Equal payment series present worth Amount

- The objective of this mode of investment is to find the present worth of an equal payment made at the end of every interest period for n interest periods at an interest rate of i compounded at the end of every interest period.



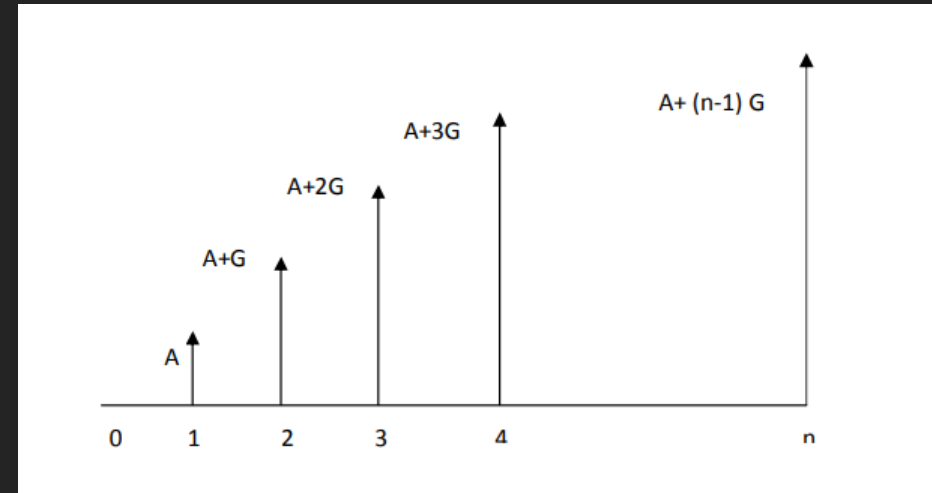
6. Equal payment series capital recovery amount

- The objective of this mode of investment is to find the annual equivalent amount (A) which is to be recovered at the end of every interest period for n interest period for a loan (P) which is sanctioned now at an interest rate of i compounded at the end of every interest period.



7. Uniform Gradient series annual Equivalent amount

- The objective of this mode of investment is to find the annual equivalent amount of a series with an amount A at the end of the first year and with an equal increment (G) at the end of each of the following $n-1$ years with an interest rate i compounded annually.





THANK YOU